

LOUIS HONORÉ LOUA

Network Engineer | Routing, Switching & Network Security

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PROFESSIONAL SUMMARY

First Class Honours Computer Science graduate specialising in network engineering, with hands-on experience configuring and securing Cisco routing and switching environments on real hardware. Skilled in VLANs and inter-VLAN routing, subnetting, ACLs, AAA (RADIUS/TACACS+), IPSec/GRE tunnelling and software-defined networking. Author of an award-winning SDN dissertation with measurable performance and security gains. Seeking a Network Engineer role to design, deploy and secure reliable enterprise networks.

TECHNICAL SKILLS

Routing & Switching: Cisco IOS, VLANs (802.1Q), inter-VLAN routing, static & dynamic routing, subnetting/VLSM, switch & router configuration

Network Security: ACLs, AAA (RADIUS, TACACS+), IPSec site-to-site VPN, GRE tunnelling, traffic isolation, network security policy enforcement

SDN & Automation: Software-Defined Networking, OpenFlow, Ryu controller, Open vSwitch, Mininet, network emulation

Systems & Scripting: Linux system administration, Python, C, C++, PowerShell, HTML

Tools: Cisco Packet Tracer, Cisco hardware labs, Wireshark (basics), data collection & analysis

PROFESSIONAL EXPERIENCE

Network & Cybersecurity Intern — TrusterLabs Ltd.

Apr – Jun 2026

Kigali, Rwanda • Supervisor: Dominique Harelimana (Cisco Advanced Instructor)

- **Configured and troubleshot** Cisco routers and switches on real hardware, including VLANs, inter-VLAN routing, IP addressing and subnetting.
- **Applied network security fundamentals** — device hardening, access control and best practices aligned with Cisco Networking Academy standards (CCNA 1 & 2, Cybersecurity Essentials).
- **Administered Linux systems** and performed network configuration and diagnostics in a professional lab environment during a competitive two-month internship required for the INES-Ruhengeri degree.

KEY PROJECTS

Automated VLAN Management using SDN — Undergraduate Dissertation

2025 – 2026

Cisco hardware • Mininet • Open vSwitch • Ryu controller

- **Designed and evaluated** a software-defined networking system automating VLAN management and directional security-policy enforcement across a campus network.
- **Achieved 89.5% latency reduction, 99.1% jitter reduction, zero packet loss** on authorised flows, and 100% automated enforcement of security policies.
- **Graded 19.5/20** — the highest dissertation score recorded university-wide that year; first-author research paper in preparation.

Cisco Security & WAN Labs (Packet Tracer)

2024 – 2026

- **AAA access control:** deployed RADIUS and TACACS+ authentication to centrally secure administrative access to network devices.
- **Secure WAN connectivity:** built IPSec site-to-site VPN tunnels and GRE tunnelling for encrypted inter-site traffic in an ISP-scale topology.

EDUCATION

BSc in Computer Science — Network Engineering

2023 – 2026

INES-Ruhengeri, Musanze, Rwanda • Medium of instruction: English

- **First Class Honours** — cumulative average 81.25% (GPA 5.0/5.0); final-year average 85.85%.
- Relevant coursework: Cryptography & System Security, WAN Technologies, Network Protocols, LAN Technologies, Machine Learning & Neural Networks, IT Project Management.

Graduate Bridge Program — Carnegie Mellon University Africa (CMU-Africa)

2026

- Completed CMU-Africa's competitive graduate Bridge Program, strengthening advanced computing and academic foundations.

CERTIFICATIONS

- **Cisco CCNA: Introduction to Networks** — Cisco Networking Academy (expected Dec 2026) • **Introduction to Cybersecurity** — Cisco Networking Academy (2026).
- **Networking & Cybersecurity Internship Certificate** — TrusterLabs Ltd. (2026).

LANGUAGES

French (native) • English (fluent — full degree completed in English) • **Susu & Kpelle** (native)